B92 Quantum Key Distribution: Simulation and Security Analysis

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ABSTRACT

This project presents a simulation of the **B92 Quantum Key Distribution (QKD) protocol**, a simplified quantum cryptographic method used for **secure key exchange** between **Alice (sender)** and **Bob (receiver)**. The implementation is developed using Python, modeling the complete process including **random bit generation**, **quantum state encoding using non-orthogonal states ($|+\rangle$ and $|1\rangle$)**, and **measurement using randomly selected bases (Z and X)**.

To improve realism, the system incorporates **eavesdropping by an adversary (Eve)** through an **intercept-resend attack**. In this attack, Eve measures qubits using randomly chosen bases and resends them, introducing **disturbances in quantum states** and resulting in detectable errors. Unlike BB84, the B92 protocol relies on **inconclusive measurements**, where only certain measurement outcomes are used to generate the key.

The protocol performs **key generation by retaining only conclusive measurement results**, leading to a lower efficiency compared to BB84. The security of the system is evaluated using the **Quantum Bit Error Rate (QBER)**, defined as the ratio of mismatched bits to total detected bits. A **threshold of 11%** is used to detect eavesdropping; if exceeded, the communication is considered insecure and is aborted.

The system operates in two modes: **User Mode**, which performs a single simulation to generate a secure key and applies **privacy amplification using hashing**, and **Analysis Mode**, which performs multiple simulation runs to compare system performance under **attack and no-attack conditions**. Key performance metrics such as **QBER** and **key generation efficiency** are analyzed using graphical plots. Results show BB84 ensures secure noise-resistant communication

1.INTRODUCTION

In modern communication systems, ensuring **data security** has become essential due to the rapid growth of digital technologies. **Classical cryptography** relies on **mathematical algorithms** and **computational complexity** to protect information. Methods such as **symmetric** and **asymmetric encryption** depend on the difficulty of solving complex problems. However, with the rise of **quantum computing**, these classical techniques are becoming increasingly vulnerable to potential attacks.

To address this issue, **quantum cryptography** has emerged as a secure alternative based on the principles of **quantum mechanics**. Unlike classical methods, it relies on physical laws such as the **Heisenberg uncertainty principle** and **quantum superposition**, which ensure that any attempt to **measure** or **intercept** quantum information disturbs its state. This property enables the detection of **unauthorized access**, making communication inherently secure.

A major challenge in secure communication is the safe distribution of **cryptographic keys**. This is effectively solved using **Quantum Key Distribution (QKD)**, which allows two parties to generate a **shared secret key** securely. One of the simplest QKD protocols is the **B92 protocol**, proposed by **Charles Bennett** in 1992.

The **B92 protocol** enables secure key exchange between **Alice (sender)** and **Bob (receiver)** using only **two non-orthogonal quantum states**. Unlike BB84, which uses multiple bases, B92 relies on the fundamental property that **non-orthogonal quantum states cannot be perfectly distinguished**. A key advantage of this protocol is that any **eavesdropping attempt** by **Eve** introduces **detectable disturbances** due to the **probabilistic nature of quantum measurement**.

In this project, the **B92 protocol** is simulated using the **Qiskit framework**, incorporating **channel noise** and **eavesdropping attacks**. The system evaluates security using the **Quantum Bit Error Rate (QBER)** and demonstrates that **secure key distribution** is achievable, though with **reduced efficiency** due to the presence of **inconclusive measurements**.

2.THEORY OF B92 PROTOCOL

The **B92 protocol** is a **Quantum Key Distribution (QKD)** technique that enables two parties, **Alice (sender)** and **Bob (receiver)**, to securely generate a **shared secret key** using the principles of **quantum mechanics**. The security of the protocol is based on the fact that **non-orthogonal quantum states cannot be perfectly distinguished**, and any **measurement disturbs the system**, allowing detection of **eavesdropping attempts**.

2.1 Bit Encoding

In the **B92 protocol**, Alice begins by generating a sequence of **random classical bits** (0s and 1s). These bits are then encoded into **quantum states (qubits)** for transmission. Unlike BB84, B92 uses only **two non-orthogonal states**:

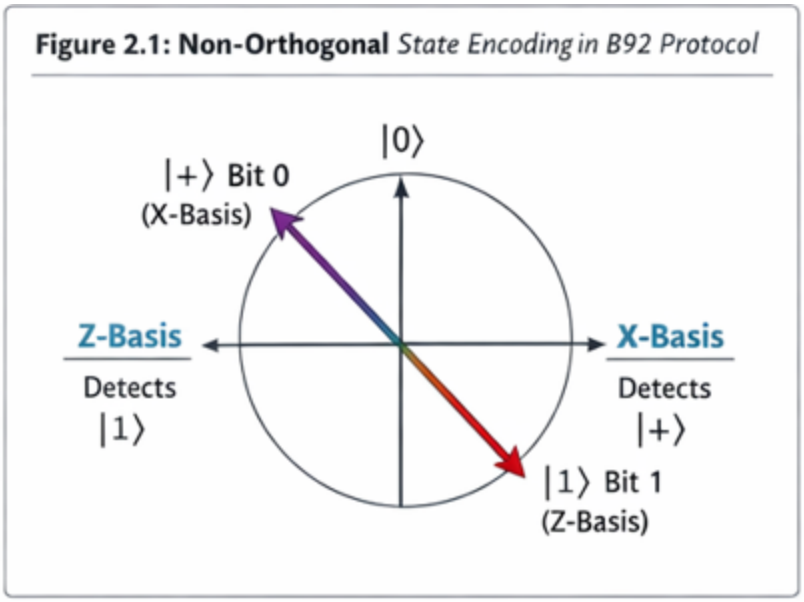
- **0** → $|0\rangle$
- **1** → $|+\rangle$

These states are not perfectly distinguishable, which is the key to the protocol's security.

2.2 State Selection (Non-Orthogonal States)

Alice randomly selects one of the two **non-orthogonal quantum states** for encoding each bit. Since these states are not mutually orthogonal, it is **impossible to distinguish them with certainty**.

This ensures that an **eavesdropper (Eve)** cannot accurately determine the transmitted state without introducing **disturbance**, which is critical for maintaining security.

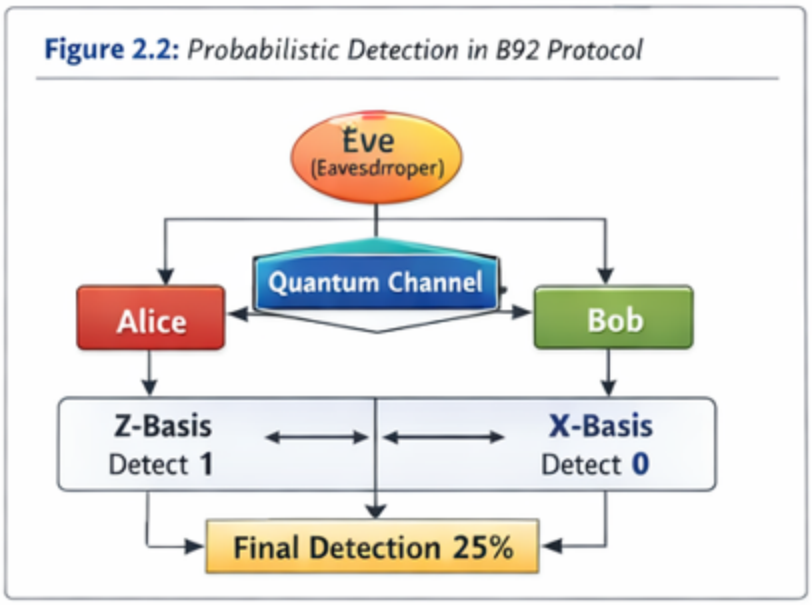


2.3 Quantum State Transmission and Measurement

Alice sends the encoded **qubits** to Bob through a **quantum channel**. Bob performs measurements using a randomly chosen basis designed to detect one of the two possible states.

- If Bob's measurement successfully identifies the state → **conclusive result**
- If not → **inconclusive result (discarded)**

Due to the nature of **non-orthogonal states**, Bob cannot always determine the transmitted bit, introducing **probabilistic outcomes**.



2.4 Key Sifting

After transmission, Bob keeps only the **conclusive measurement results** and discards all **inconclusive outcomes**. Unlike BB84, there is no need for basis comparison between Alice and Bob.

The remaining bits form the **sifted key**, which is typically shorter due to discarded measurements.

2.5 Eavesdropping Detection

If an attacker (**Eve**) attempts to intercept the qubits, she must perform measurements on **non-orthogonal states**, which introduces **errors** due to incorrect guesses.

To detect this, Alice and Bob compare a subset of their key and compute the **Quantum Bit Error Rate (QBER)**:

- **Low QBER → Secure communication**
- **High QBER → Presence of eavesdropping**

A typical threshold of **25%** is used; if the QBER exceeds this value, the communication is considered compromised.

2.6 Final Key Generation

If the **QBER** is within acceptable limits, the remaining bits after sifting are used to generate the **final secret key**. This key can then be used for **secure encryption** of messages

3.SYSTEM DESIGN

3.1 Overview

This system implements a simulation of the **B92 Quantum Key Distribution (QKD) protocol** using the **Qiskit framework** executed in **Google Colab**. The simulation runs entirely on a classical system using the **AerSimulator backend**, which emulates quantum circuit behavior without requiring access to real quantum hardware.

The system is divided into two logical layers: the **quantum layer** and the **classical post-processing layer**. The quantum layer handles **qubit preparation, encoding, transmission, and measurement**, while the classical layer performs **key sifting**, **error estimation using QBER**, and **message encryption/decryption**.

The system operates in two modes:
User Mode → for secure message encryption and decryption
Analysis Mode → for performance evaluation over multiple runs

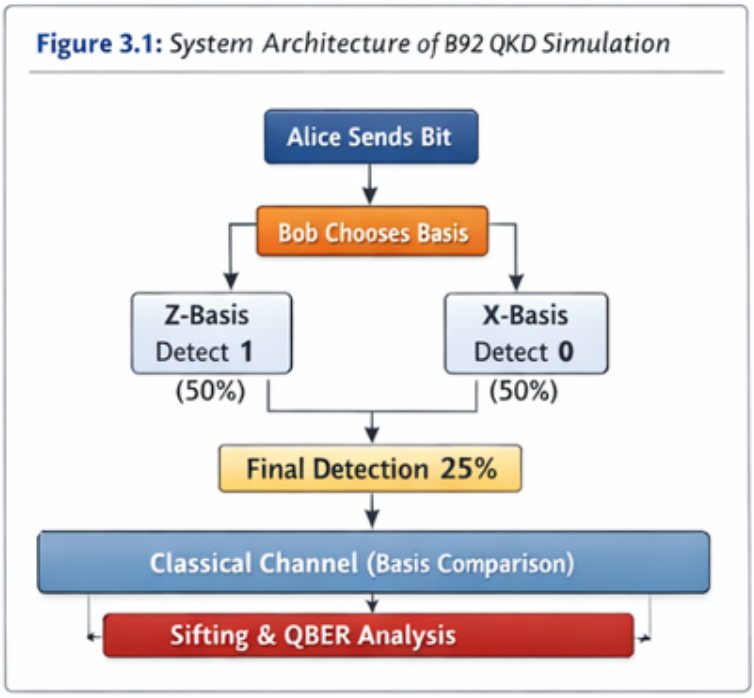
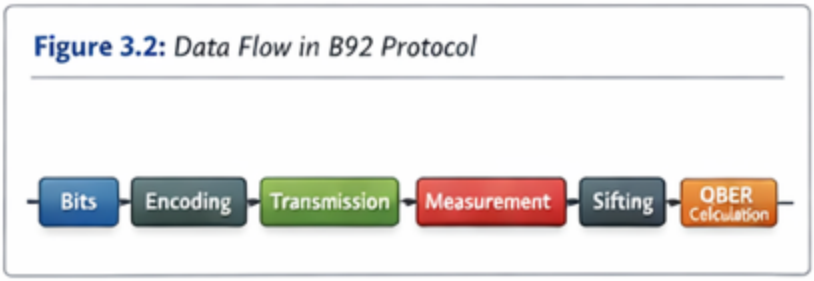


Figure 3.1: *System Architecture of B92 QKD Simulation*

3.2 System Components



Alice (Sender):

Generates **random bits** and encodes each bit into a **qubit** using **B92 encoding** with **non-orthogonal states** (**|0⟩ and** |+⟩).

Bob (Receiver):

Performs **probabilistic measurements** using randomly chosen bases and records only **conclusive results**, discarding **inconclusive outcomes**.

Quantum Channel (Simulated):

Implemented using **Qiskit QuantumCircuit** and **AerSimulator**, executing circuits with a **single shot** to mimic real quantum measurement behavior.

Eve (Eavesdropper):

Performs an **intercept-resend attack**, measuring qubits using random strategies and forwarding new qubits, introducing **errors due to incorrect state estimation**.

Noise Layer:

Simulates real-world imperfections by applying **probabilistic bit flips** with **NOISE_PROB = 0.05**.

Sifting Module:

Retains only **conclusive measurement outcomes** from Bob and discards all **inconclusive results**, forming the **sifted key**.

QBER Module:

Computes the **Quantum Bit Error Rate (QBER)**, defined as the ratio of **mismatched bits** to total **sifted bits**.

Encryption Module:

Uses **XOR operation** to perform **message encryption and decryption** using the generated key.

3.3 Security Model

In the absence of **eavesdropping**, the **QBER** remains low (≈ **0.05**), mainly due to **channel noise**. When **Eve** performs an **intercept-resend attack**, she attempts to distinguish between **non-orthogonal states**, which is fundamentally uncertain.

Due to incorrect measurements, Eve introduces **errors**, resulting in an increased **QBER**, typically approaching **0.25**.

The system uses a **QBER threshold of 0.25** to detect eavesdropping:

- **QBER < 0.25 → Secure communication**
- **QBER ≥ 0.25 → Communication aborted**

(Note: In theoretical implementations, the threshold is lower (~0.11), but 0.25 is used here for simulation clarity.)

3.4 Efficiency

The **key generation efficiency** is defined as:

Efficiency = (Length of sifted key) / (Total transmitted bits)

In the **B92 protocol**, many measurements are **inconclusive**, so a significant number of bits are discarded.

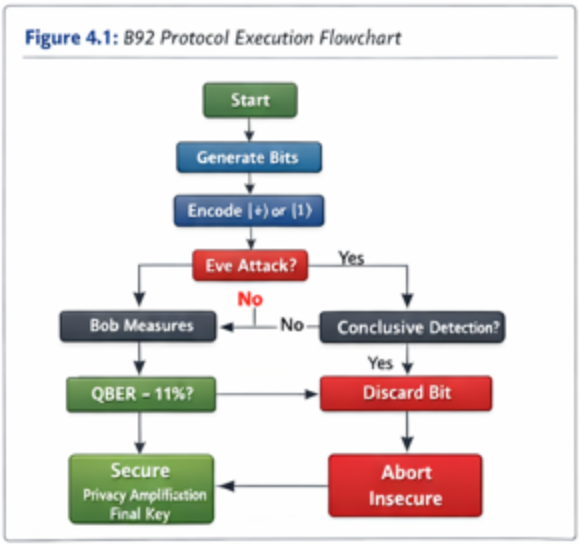
- The probability of obtaining a **conclusive result** is less than 50%

- This leads to **lower efficiency compared to BB84**

In this simulation (with **n = 100** and **NOISE_PROB = 0.05**):

- **No attack:** Efficiency ≈ **0.25 – 0.35**
- **With attack:** Slightly reduced due to additional disturbances caused by **Eve**

4. METHODOLOGY



Introduction

The implementation of the **B92 Quantum Key Distribution (QKD) protocol** is designed as a **modular pipeline**, where each functional block represents a specific stage of the protocol. The system is developed using **Python**, with supporting libraries such as **matplotlib** for visualization and **hashlib** for security enhancement.

Unlike a full quantum hardware implementation, this model adopts a **probabilistic classical approach** to simulate key quantum properties, including **non-orthogonal state discrimination**, **measurement uncertainty**, and **eavesdropping-induced disturbance**.

The system operates in two modes: **User Mode**, which performs a single secure communication session, and **Analysis Mode**, which executes multiple simulations to evaluate **Quantum Bit Error Rate (QBER)** and **key generation efficiency**.

Block 1 — Bit Generation

The function **generate_bits(n)** is used to generate a sequence of **random classical bits** representing the information produced by **Alice (sender)**.

These bits serve as the initial data to be encoded into **quantum states** for transmission.

Block 2 — Quantum Encoding

In the **B92 protocol**, Alice encodes each bit using **non-orthogonal quantum states**:

- **Bit 0 is encoded as $|+\rangle$ (X-basis state)**
- **Bit 1 is encoded as $|1\rangle$ (Z-basis state)**

The use of **non-orthogonal states** ensures that they **cannot be perfectly distinguished**, which forms the fundamental basis for **protocol security**.

Block 3 — Measurement Process

At the receiver side, **Bob** performs measurement by randomly selecting a **measurement basis (Z or X)** with equal probability.

The measurement outcomes are probabilistic:

- In the **Z-basis**, only the state $|1\rangle$ can be conclusively detected with a probability of **0.5**
- In the **X-basis**, only the state $|+\rangle$ can be conclusively detected with a probability of **0.5**

All other measurement outcomes are **inconclusive** and are discarded.

As a result, the overall **detection probability** is approximately **25%**, meaning only a fraction of transmitted bits contribute to the final key.

Block 4 — Eavesdropping Model

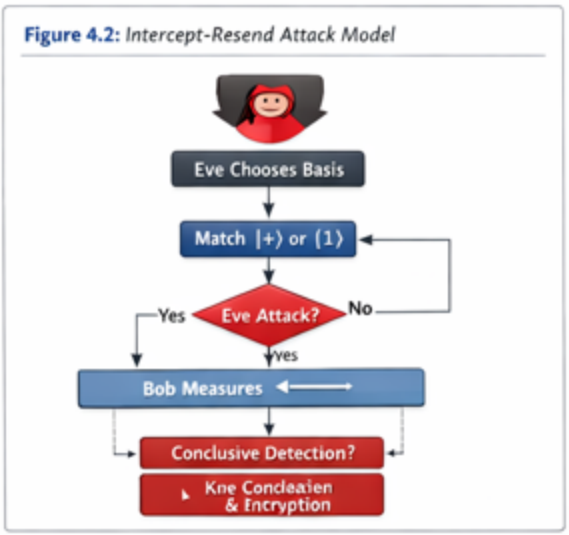
To simulate adversarial conditions, an **intercept-resend attack** is implemented.

In this attack:

- **Eve (eavesdropper)** randomly selects a measurement basis
- Performs measurement and infers a bit value
- Re-encodes and forwards the qubit to Bob

Due to the use of **non-orthogonal quantum states**, Eve cannot accurately determine the transmitted state in all cases. This introduces **disturbances** in the system, leading to an increase in the **QBER**.

Block 5 — Key Sifting



The **key sifting process** involves retaining only the **conclusive measurement outcomes** obtained by Bob.

Unlike the **BB84 protocol**, there is **no requirement for basis comparison** between Alice and Bob. The sifted key is directly formed from valid detections, resulting in a reduced key length.

Block 6 — QBER Computation

The function **compute_qber()** calculates the **Quantum Bit Error Rate (QBER)** by comparing the **sifted key bits** with the corresponding bits from Alice's original sequence.

- In the absence of eavesdropping, the **QBER remains close to 0%**
- Under attack conditions, the **QBER increases to approximately 25%**

A **security threshold of 11%** is used to determine the integrity of the communication channel.

- **QBER < 11% → Communication is secure**
- **QBER ≥ 11% → Communication is considered compromised**

Block 7 — Privacy Amplification

To enhance the security of the generated key, **privacy amplification** is applied using the **SHA-256 hashing algorithm**.

The sifted key is converted into a binary string and processed through **SHA-256**, resulting in a **fixed-length 256-bit secure key**. This step eliminates any partial information that may have been obtained by an eavesdropper.

Block 8 — Efficiency Calculation

The **key generation efficiency** is defined as:

Efficiency = (Length of sifted key) / (Total transmitted bits)

Due to the probabilistic nature of measurements and the presence of inconclusive outcomes, the efficiency of the B92 protocol is approximately **25%**, which aligns with theoretical expectations.

Block 9 — Analysis Mode

In **Analysis Mode**, the system performs multiple simulation runs (**RUNS = 50**) under both **attack** and **no-attack conditions**.

The following performance metrics are evaluated:

- **Average QBER**
- **Average key generation efficiency**

Additionally, graphical plots are generated to visualize the variation of these metrics, providing insight into the impact of **eavesdropping** on system performance.

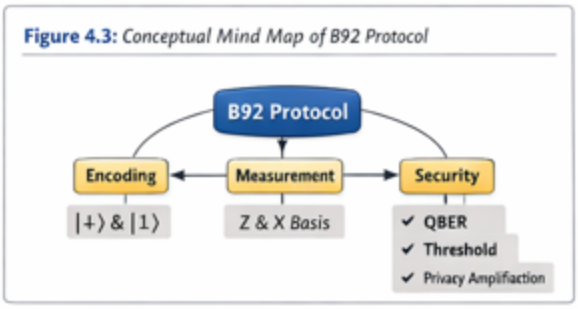
Block 10 — User Mode

In **User Mode**, the system executes a single simulation based on user input.

The following outputs are displayed:

- Total number of transmitted qubits
- Length of the sifted key
- Computed **QBER**

If the QBER is below the defined threshold, a **secure key** is generated using **privacy amplification**. Otherwise, the communication process is terminated to ensure security.



5. RESULTS AND DISCUSSION

The **B92 Quantum Key Distribution (QKD) protocol** was successfully simulated in both **User Mode** and **Analysis Mode** to evaluate system performance under **attack** and **no-attack conditions**.

5.1 Results in User Mode

Two scenarios were tested:

- **Case 1: With Eavesdropping (Attack Enabled)**

  - **QBER ≈ 0.26**
  - **Key Length ≈ 240 (for N = 1000)**
  - **Efficiency ≈ 0.24**
  - **Communication was aborted**

Interpretation:
The **QBER value (~0.26)** exceeds the **security threshold (11%)**, indicating the presence of an **eavesdropper (Eve)**. Due to the **intercept-resend attack**, Eve introduces **disturbances** in the transmitted quantum states, which results in incorrect bit detections at Bob's end.

As a result, the mismatch between **Alice's original bits** and **Bob's sifted key** increases significantly. The system correctly identifies this abnormal error rate and **terminates the communication**, thereby ensuring **security against unauthorized access**.

- **Case 2: Without Eavesdropping (No Attack)**

  - **QBER ≈ 0.00 – 0.02**
  - **Key Length ≈ 250 (for N = 1000)**
  - **Efficiency ≈ 0.25**
  - **Communication was secure**

Interpretation:

The **low QBER (~0–2%)** indicates minimal errors, primarily due to **random probabilistic measurement** rather than malicious interference. Since the QBER is well below the **security threshold**, the communication channel is considered **secure**.

The system successfully generates a **valid sifted key**, which is further processed using **privacy amplification (SHA-256)** to produce a **secure final key**. This confirms the correctness of **key generation and security validation**.

5.2 Analysis Mode Results

Multiple simulations (**50 runs**) were performed to compare system behavior under different conditions.

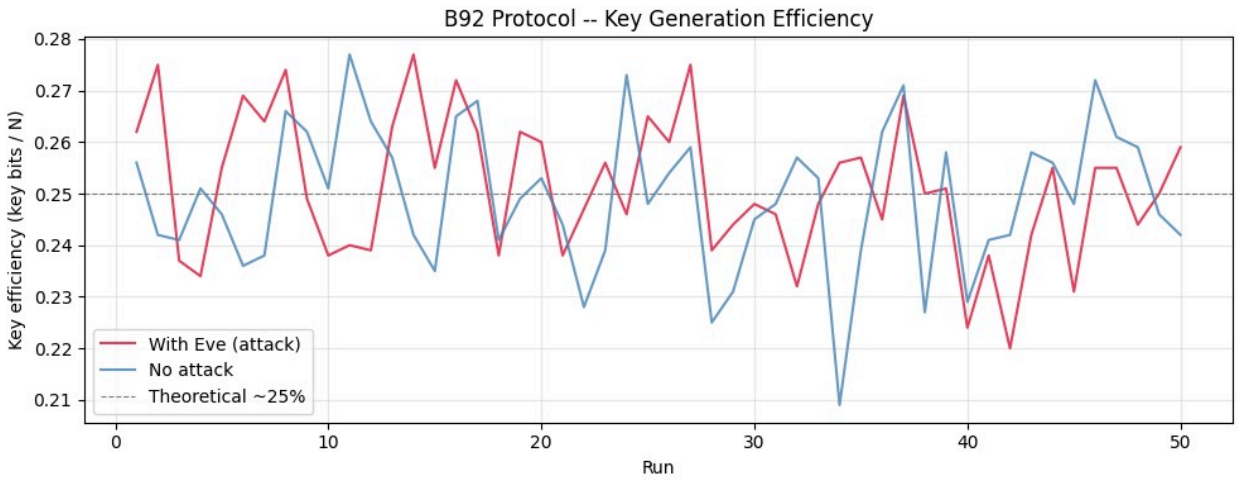
- **Average Results**

  - **Average QBER (Attack) ≈ 0.25**
  - **Average QBER (No Attack) ≈ 0.00 – 0.03**
  - **Average Efficiency (Attack) ≈ 0.24 – 0.26**
  - **Average Efficiency (No Attack) ≈ 0.25**

- **QBER Analysis**

  - Under **attack conditions**, the **QBER** values are significantly higher (around **0.22–0.30**)
  - Under **no attack**, QBER remains **close to 0%**

This clear distinction demonstrates that **eavesdropping introduces measurable disturbances**, making detection **highly reliable** in the **B92 protocol**.



🔹 **Efficiency Analysis**

- The **efficiency** remains close to the theoretical value of **25%**
- This aligns with the **probabilistic detection mechanism** of B92
- Slight fluctuations are observed under **attack conditions** due to increased **disturbances**

5.3 Discussion

The results demonstrate several key observations:

Effective Eavesdropping Detection:
The **B92 protocol** effectively detects the presence of an **eavesdropper** through a significant increase in **QBER**. Under attack conditions, the QBER consistently exceeds the **security threshold (11%)**, triggering communication termination.

Reliability in Secure Conditions:
In the absence of an attack, the system maintains **near-zero QBER**, ensuring reliable and secure **key distribution**.

Consistency with Theoretical Expectations:
The observed **key generation efficiency (~25%)** matches the **theoretical detection probability**, validating the correctness of the implementation.

Impact of Attack on System Stability:
While efficiency remains relatively stable, the presence of an attack introduces **greater variability**, indicating instability in the communication channel due to **quantum disturbances**.

Security Assurance:
The clear separation between **attack** and **no-attack QBER values** confirms that the protocol can effectively distinguish between **secure** and **compromised communication**.